

Introduction to Statistical Methods
(S1-25 AIMLCLZC418) – Assignment 2
AIML Section - 4

Each question carries 05 Marks (5 x 2 = 10 Marks)

Duration: 25th January 2026 – 08th February 2026

- 1) Submissions are individual**
- 2) Solve these on paper, scan, and upload**
- 3) Plagiarism results in zero marks**
- 4) Write your name, BITS ID and Section on each page**
- 5) AI generated graphs/diagrams result in zero marks.**
- 6) Please make sure that you show all working clearly, state any assumptions made, use appropriate statistical notation, and interpret results in the context of the problem.**

Problem-1

A factory produces metal rods with lengths that vary slightly around a target value of 50 cm. To study the behavior of the sampling distribution of the sample mean, a quality control engineer collects 10 random samples, each consisting of 6 rods, from the production process.

Assume that the population of rod lengths has a known mean of 50 cm and a known standard deviation of 1 cm, and that all observations are independent and randomly selected.

The observed rod lengths (in cm) for each of the 10 samples are given below.

Sample data (in cm)

Sample Rod Lengths ($n = 6$)

Sample 1: 49, 50, 51, 50, 49, 51

Sample 2: 50, 51, 49, 50, 50, 50

Sample 3: 51, 50, 49, 50, 51, 49

Sample 4: 48, 49, 50, 50, 51, 52

Sample 5: 50, 50, 51, 49, 50, 50

Sample 6: 49, 50, 50, 51, 50, 50

Sample 7: 50, 49, 51, 50, 49, 51

Sample 8: 51, 50, 50, 49, 50, 50

Sample 9: 49, 50, 51, 50, 51, 49

Sample 10: 50, 50, 49, 51, 50, 50

Using the above data,

- a)** Compute the sample mean for each of the 10 samples.
- b)** Using the sample means obtained in part (a), calculate the mean and standard deviation of the sampling distribution of the sample mean.
- c)** Calculate the probability that a randomly selected sample mean exceeds 50.5 cm.
- d)** Calculate the probability that a randomly selected sample mean falls below 49.5 cm.
- e)** Based on parts (c) and (d), decide whether such sample means are rare or common under normal operating conditions. Interpret what this outcome implies about the stability of the production process.
- f)** Construct a histogram and a dot plot of the 10 sample means.
- g)** If the engineer raises the sample size from 6 to 20 rods per sample, explain how the spread of the sampling distribution would change.

Problem-2

A factory produces small parts whose lengths are supposed to be 50 units. To monitor the process, a quality control analyst randomly selects 24 parts and measures their lengths (in units). The measurements are:

49, 50, 51, 50, 50, 49, 50, 51, 50, 50, 49, 50, 50, 51, 50, 50, 49, 50, 51, 50, 50, 50, 49, 50

Population mean = 50 units.

- a)** Compute the sample mean and sample standard deviation.
- b)** Build a 90% confidence interval for the population mean. Illustrate the confidence interval by plotting it on a graph, clearly marking the sample mean, lower bound, and upper bound.
- c)** Based on the confidence interval, decide whether the production process meets the target population mean.
- d)** Explain how the width of the confidence interval would change if:
 - i) the sample size were increased, or
 - ii) the confidence level were raised to 98%.
- e)** If the lower limit of the confidence interval falls below 49 units, recommend steps the quality control team should take.
- f)** If the company wants a 99% confidence interval for the population mean (still assuming population mean = 50 units), analyze how the required sample size would change and justify why.

*****Happy Learning*****